Mini Project: Impact of Air Pollution on Health and Environment in Gudipallipadu

Community Service Project, Sree Venkateswara College of Engineering.

Run every cell from top to bottom.

Step 1: Install the required libraries

In [1]:

```
import sys, subprocess
subprocess.check_call([sys.executable, "-m", "pip", "install", "reportlab", "matplotlib", "pillow"
      ])
print("Libraries ready")
```

Output:

```
Requirement already satisfied: reportlab in /usr/local/lib/python3.12/dist-packages (5.0.1)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.11.2)
Requirement already satisfied: pillow in /usr/local/lib/python3.12/dist-packages (13.0.0)
Requirement already satisfied: charset-normalizer in /usr/local/lib/python3.12/dist-packages (from
 reportlab) (3.5.2)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from m
atplotlib) (1.4.0)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matpl
otlib) (0.12.1)
Requirement already satisfied: fonttools>=4.28.2 in /usr/local/lib/python3.12/dist-packages (from
matplotlib) (4.67.0)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from
matplotlib) (1.5.1)
Requirement already satisfied: numpy>=1.25 in /usr/local/lib/python3.12/dist-packages (from matplo
tlib) (2.5.4)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from ma
tplotlib) (26.3)
Requirement already satisfied: pyparsing>=3 in /usr/local/lib/python3.12/dist-packages (from matpl
otlib) (3.3.3)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (fr
om matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-da
teutil>=2.7->matplotlib) (1.17.0)
Libraries ready
```

Step 2: Survey data (100 residents)

Edit only the SURVEY dictionary if your real numbers differ.

In [2]:

```
import os

OUT = "output"
CH = os.path.join(OUT, "charts")
os.makedirs(CH, exist_ok=True)

# ---------------------------------------------------------------------
# 1. SURVEY DATA  (counts out of TOTAL residents)  -> edit if needed
# ---------------------------------------------------------------------
TOTAL = 100
SURVEY = {
    "aware": 75,
    "not_aware": 25,
    "health_problems": 40,
    "respiratory": 40,
    "cardiovascular": 20,
    "income_agriculture": 50,
    "income_daily_wage": 30,
    "income_other": 20,
    "economic_loss": 20,
}
```

Step 3: Analysis and output

In [3]:

```
# ---------------------------------------------------------------------
# 2. ANALYSIS + CONSOLE OUTPUT
# ---------------------------------------------------------------------
def pct(n): return round(n * 100 / TOTAL)

def run_analysis():
    assert SURVEY["aware"] + SURVEY["not_aware"] == TOTAL, "Awareness counts must add up to TOTAL"
    assert SURVEY["income_agriculture"] + SURVEY["income_daily_wage"] + SURVEY["income_other"] ==
      TOTAL
    print("=" * 64)
    print(" MINI PROJECT - AIR POLLUTION SURVEY, GUDIPALLIPADU")
    print(f" Residents surveyed: {TOTAL}")
    print("=" * 64)
    print(f"{'Parameter':<46}{'Count':>6}{'%':>8}")
    print("-" * 64)
    labels = {
        "aware": "Aware of causes and effects",
        "not_aware": "Not aware",
        "health_problems": "Reported health problems",
        "respiratory": "Reported respiratory problems",
        "cardiovascular": "Reported cardiovascular problems",
        "income_agriculture": "Income: agriculture",
        "income_daily_wage": "Income: daily wage labour",
        "income_other": "Income: other sources",
        "economic_loss": "Reported economic loss",
    }
    for k, name in labels.items():
        print(f"{name:<46}{SURVEY[k]:>6}{pct(SURVEY[k]):>7}%")
    print("-" * 64)
    print("Conclusion: awareness is high (%d%%) but %d%% still report health problems." %
          (pct(SURVEY["aware"]), pct(SURVEY["health_problems"])))
    print("=" * 64)

run_analysis()
```

Output:

```
================================================================
 MINI PROJECT - AIR POLLUTION SURVEY, GUDIPALLIPADU
 Residents surveyed: 100
================================================================
Parameter                                       Count       %
----------------------------------------------------------------
Aware of causes and effects                        75     75%
Not aware                                          25     25%
Reported health problems                           40     40%
Reported respiratory problems                      40     40%
Reported cardiovascular problems                   20     20%
Income: agriculture                                50     50%
Income: daily wage labour                          30     30%
Income: other sources                              20     20%
Reported economic loss                             20     20%
----------------------------------------------------------------
Conclusion: awareness is high (75%) but 40% still report health problems.
================================================================
```

Step 4: Charts

In [4]:

```
# ---------------------------------------------------------------------
# 3. CHARTS
# ---------------------------------------------------------------------
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt

def make_charts():
    NAVY="#1F3A5F"; TEAL="#2A9D8F"; AMBER="#E9A23B"; RED="#D1495B"; GREY="#C9D1DA"
    plt.rcParams.update({"font.family":"DejaVu Sans","font.size":10})

    def save(fig,name):
        fig.savefig(os.path.join(CH,name),dpi=200,bbox_inches="tight"); plt.close(fig)

    # 1 awareness
    fig,ax=plt.subplots(figsize=(4.2,3.6))
```

```
    ax.pie([SURVEY["aware"],SURVEY["not_aware"]],labels=[f"Aware ({SURVEY['aware']})",f"Not aware
      ({SURVEY['not_aware']})"],colors=[TEAL,GREY],autopct="%1.0f%%",
           startangle=90,wedgeprops=dict(width=0.45,edgecolor="white"),textprops={"fontsize":10})
    ax.set_title("Awareness of causes and effects\n(n = 100) residents",fontsize=11,color=NAVY)
    save(fig,"c1.png")

    # 2 health effects
    fig,ax=plt.subplots(figsize=(5.2,3.4))
    labels=["Respiratory\nproblems","Cardiovascular\nproblems"]; vals=[pct(SURVEY["respiratory"]),
      pct(SURVEY["cardiovascular"])]
    b=ax.bar(labels,vals,color=[RED,AMBER],width=0.5)
    for r,v in zip(b,vals): ax.text(r.get_x()+r.get_width()/2,v+1.2,f"{v}%",ha="center",fontweight
      ="bold")
    ax.set_ylim(0,55); ax.set_ylabel("% of residents reporting")
    ax.set_title("Health effects reported by residents",fontsize=11,color=NAVY)
    for s in ("top","right"): ax.spines[s].set_visible(False)
    save(fig,"c2.png")

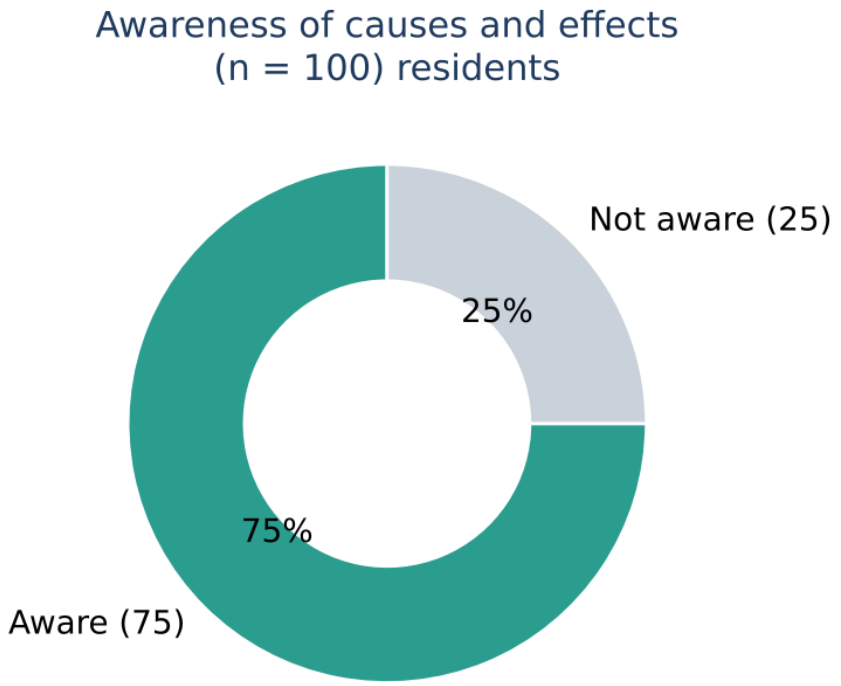
    # 3 income
    fig,ax=plt.subplots(figsize=(5.2,3.4))
    labels=["Agriculture","Daily wage\nlabour","Other sources"]; vals=[pct(SURVEY["income_agricult
      ure"]),pct(SURVEY["income_daily_wage"]),pct(SURVEY["income_other"])]
    b=ax.barh(labels[::-1],vals[::-1],color=[GREY,AMBER,TEAL],height=0.5)
    for r,v in zip(b,vals[::-1]): ax.text(v+1,r.get_y()+r.get_height()/2,f"{v}%",va="center",fontw
      eight="bold")
    ax.set_xlim(0,65); ax.set_xlabel("% of residents")
    ax.set_title("Main source of income",fontsize=11,color=NAVY)
    for s in ("top","right"): ax.spines[s].set_visible(False)
    save(fig,"c3.png")

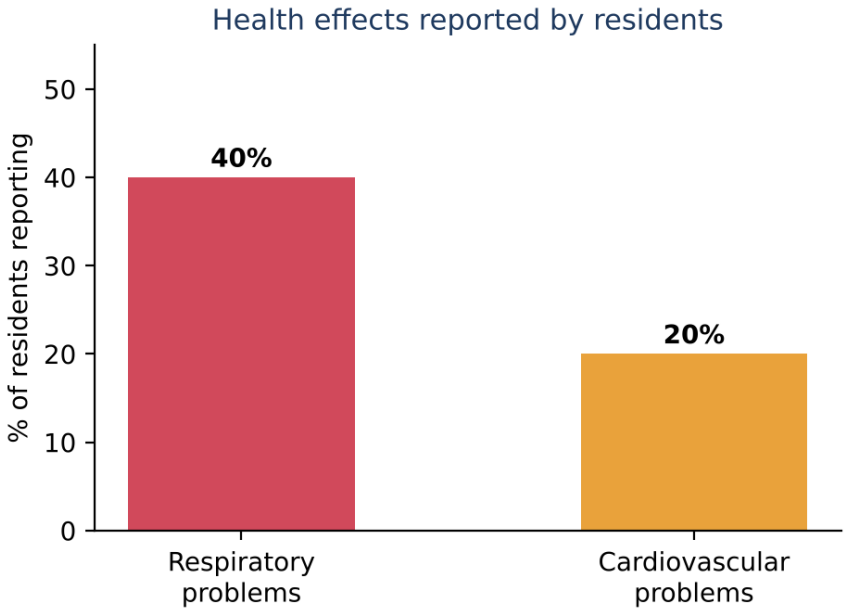
    # 4 summary
    fig,ax=plt.subplots(figsize=(6.4,3.4))
    labels=["Aware of causes\n& effects","Health problems\nreported","Respiratory\nproblems","Card
      iovascular\nproblems","Economic loss\nreported"]
    vals=[pct(SURVEY["aware"]),pct(SURVEY["health_problems"]),pct(SURVEY["respiratory"]),pct(SURVE
      Y["cardiovascular"]),pct(SURVEY["economic_loss"])]; cols=[TEAL,RED,RED,AMBER,NAVY]
    b=ax.bar(labels,vals,color=cols,width=0.55)
    for r,v in zip(b,vals): ax.text(r.get_x()+r.get_width()/2,v+1.5,f"{v}%",ha="center",fontweight
      ="bold")
    ax.set_ylim(0,90); ax.set_ylabel("% of residents (n = 100)")
    ax.set_title("Summary of survey outcomes",fontsize=11,color=NAVY)
    ax.tick_params(axis="x",labelsize=8)
    for s in ("top","right"): ax.spines[s].set_visible(False)
    save(fig,"c4.png")

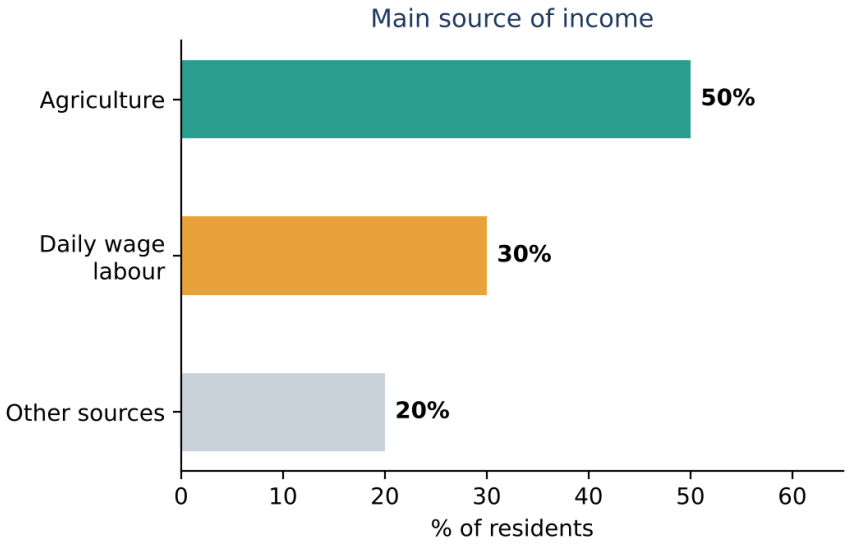
make_charts()

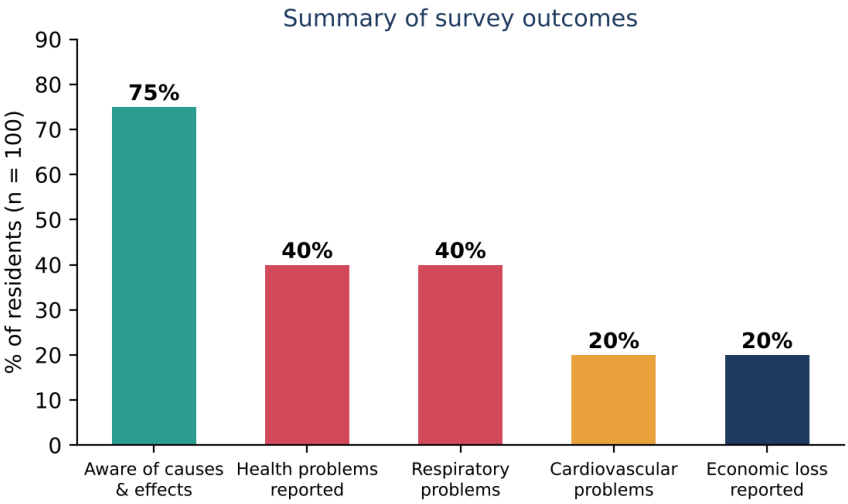
from IPython.display import Image, display
for n in ("c1", "c2", "c3", "c4"):
    display(Image(os.path.join(CH, n + ".png"), width=420))
```

Out:


Awareness of causes and effects
(n = 100) residents
Not aware (25)
25%
75%
Aware (75)


Health effects reported by residents
% of residents reporting
0
10
20
30
40
50
40%
20%
Respiratory problems
Cardiovascular problems





Step 5: Build the PDF report

In [5]:

```
# ------------------------------------------------------------------------
# 4. PDF REPORT
# ------------------------------------------------------------------------
from reportlab.lib.pagesizes import A4
from reportlab.lib.styles import ParagraphStyle
from reportlab.lib.units import cm
from reportlab.lib import colors
from reportlab.lib.enums import TA_CENTER, TA_JUSTIFY
from reportlab.platypus import (BaseDocTemplate, PageTemplate, Frame, Paragraph, Spacer,
                                PageBreak, Table, TableStyle, Image, KeepTogether, ListFlowable, L
      istItem)

NAVY=colors.HexColor("#1F3A5F"); TEAL=colors.HexColor("#2A9D8F"); LIGHT=colors.HexColor("#EEF3F8")
def build_pdf():
    global W,H,body,h1,h2,cen,cap,cell,cellb,cellh,doc,S
    W,H=A4
    body=ParagraphStyle("body",fontName="Times-Roman",fontSize=11.5,leading=17,alignment=TA_JUSTIF
      Y,spaceAfter=6)
    h1=ParagraphStyle("h1",fontName="Times-Bold",fontSize=16,leading=20,textColor=NAVY,spaceBefore
      =4,spaceAfter=8)
    h2=ParagraphStyle("h2",fontName="Times-Bold",fontSize=13,leading=17,textColor=NAVY,spaceBefore
      =8,spaceAfter=4)
    cen=ParagraphStyle("cen",parent=body,alignment=TA_CENTER)
    cap=ParagraphStyle("cap",fontName="Times-Italic",fontSize=10,leading=13,alignment=TA_CENTER,te
      xtColor=colors.HexColor("#444444"),spaceAfter=10)
    cell=ParagraphStyle("cell",fontName="Times-Roman",fontSize=10.5,leading=14)
    cellb=ParagraphStyle("cellb",parent=cell,fontName="Times-Bold")
    cellh=ParagraphStyle("cellh",parent=cell,fontName="Times-Bold",textColor=colors.white)
```

```
def P(t,s=body): return Paragraph(t,s)
def bullets(items):
    return ListFlowable([ListItem(P(i),leftIndent=14) for i in items],bulletType="bullet",star
  t="•",leftIndent=16,bulletFontSize=10)
def tbl(rows,widths,header=True):
    data=[[Paragraph(str(c),cellh if (header and r==0) else cell) for c in row] for r,row in e
  numerate(rows)]
    t=Table(data,colWidths=widths,repeatRows=1 if header else 0)
    st=[("GRID",(0,0),(-1,-1),0.5,colors.HexColor("#9AA8B8")),("VALIGN",(0,0),(-1,-1),"MIDDLE"
  ),
        ("TOPPADDING",(0,0),(-1,-1),5),("BOTTOMPADDING",(0,0),(-1,-1),5)]
    if header: st.append(("BACKGROUND",(0,0),(-1,0),NAVY))
    for r in range(1,len(rows)):
        if r%2==0: st.append(("BACKGROUND",(0,r),(-1,r),LIGHT))
    t.setStyle(TableStyle(st)); return t
def img(path,w):
    from PIL import Image as PI
    iw,ih=PI.open(path).size
    return Image(path,width=w,height=w*ih/iw)

def footer(c,d):
    if d.page>1:
        c.saveState(); c.setFont("Times-Roman",9); c.setFillColor(colors.HexColor("#555555"))
        c.drawString(2.2*cm,1.2*cm,"Mini Project: Impact of Air Pollution on Health and Enviro
  nment in Gudipallipadu")
        c.drawRightString(W-2.2*cm,1.2*cm,f"Page {d.page}")
        c.setStrokeColor(colors.HexColor("#B8C3CF")); c.line(2.2*cm,1.6*cm,W-2.2*cm,1.6*cm); c
  .restoreState()

doc=BaseDocTemplate(os.path.join("output","Mini_Project_Report_Air_Pollution.pdf"),pagesize=A4
  ,
    leftMargin=2.2*cm,rightMargin=2.2*cm,topMargin=2.2*cm,bottomMargin=2.2*cm,
    title="Mini Project Report - Impact of Air Pollution on Health and Environment in Gudipall
  ipadu",
    author="Komirika Vennela")
doc.addPageTemplates([PageTemplate(id="p",frames=[Frame(doc.leftMargin,doc.bottomMargin,doc.wi
  dth,doc.height,id="f")],onPage=footer)])
S=[]

# ---------- Cover ----------
big=ParagraphStyle("big",fontName="Times-Bold",fontSize=20,leading=26,alignment=TA_CENTER,text
  Color=NAVY)
S+= [Spacer(1,0.6*cm),P("<b>AP STATE COUNCIL OF HIGHER EDUCATION</b>",cen),
     P("(A STATUTORY BODY OF GOVERNMENT OF ANDHRA PRADESH)",cen),Spacer(1,0.8*cm),
     P("<b>COMMUNITY SERVICE PROJECT</b>",cen),P("<b>MINI PROJECT REPORT</b>",cen),Spacer(1,0.
  8*cm),
     P("on",cen),Spacer(1,0.3*cm),
     Paragraph("IMPACT OF AIR POLLUTION ON HEALTH AND ENVIRONMENT IN GUDIPALLIPADU",big),
     Spacer(1,0.8*cm),P("By",cen),P("<b>KOMIRIKA VENNELA</b>",cen),P("<b>Reg. No. 24JNA05355</
  b>",cen),Spacer(1,0.5*cm),
     P("Under the guidance of",cen),P("<b>Ms. Ch. VANAJA, M.Tech, Assistant Professor</b>",cen
  ),
     P("Department of Computer Science and Engineering",cen),Spacer(1,1.0*cm),
     P("<b>SREE VENKATESWARA COLLEGE OF ENGINEERING</b>",cen),
     P("(Accredited with NBA and NAAC 'A' grade, UGC 2(f) Recognized &amp;amp; ISO 9001:2015 Certi
  fied)<br/>"
       "(Approved by AICTE, New Delhi and Affiliated to JNTUA, Anantapuramu)<br/>"
       "North Rajupalem, Kodavaluru (V&amp;amp;M), S.P.S.R Nellore (Dt) - 524316",cen),
     Spacer(1,0.6*cm),P("<b>2024 - 2028</b>",cen),PageBreak()]

# ---------- Project details ----------
S+=[P("Project Details",h1),
 tbl([["Name of the Student","KOMIRIKA VENNELA"],["Registration Number","24JNA05355"],
      ["Program / Year","B.Tech (Computer Science and Engineering), III Year"],
      ["College","Sree Venkateswara College of Engineering"],
      ["Faculty Guide","Ms. Ch. Vanaja, M.Tech, Assistant Professor"],
      ["Community / Habitation","Gudipallipadu"],
      ["Mini Project Title","Impact of Air Pollution on Health and Environment in Gudipallipad
  u"],
      ["Mini Project Period","13-07-2026 to 26-07-2026 (2 weeks)"],
      ["Main Project Period","01-06-2026 to 12-07-2026 (6 weeks)"]],[5.2*cm,11.4*cm],header=Fa
  lse),
 Spacer(1,0.6*cm),P("Contents",h1),
 tbl([["1","Introduction"],["2","Objectives"],["3","Methodology"],["4","Questionnaire"],["5","
  Data Analysis and Results"],
      ["6","Findings"],["7","Discussion"],["8","Recommendations"],["9","Conclusion, Limitation
  s and Future Scope"],["10","References"]],
```

```
        [1.5*cm,15.1*cm],header=False),PageBreak()]

    # ---------- 1-3 ----------
    S+=[P("1. Introduction",h1),
     P("Air pollution is the presence of harmful substances in the air at levels that can damage h
      uman health, "
       "the environment and the climate. Common sources include vehicle emissions, industrial acti
      vity, agricultural "
       "practices, burning of waste and natural events such as dust storms. Exposure is linked to
      respiratory illness, "
       "cardiovascular disease and wider ecosystem damage."),
     P("During the Community Service Project, awareness campaigns, school workshops and community
      meetings were conducted "
       "in Gudipallipadu. This mini project builds on that work. It studies how aware residents ar
      e of air pollution, "
       "what health effects they report, and what measures they suggest, using a questionnaire-bas
      ed survey of 100 residents."),
     P("2. Objectives",h1),
     bullets(["To assess the level of awareness about air pollution among residents.",
              "To identify the health effects of air pollution reported by residents.",
              "To suggest measures to mitigate air pollution in the community."]),
     P("3. Methodology",h1),
     tbl([["Step","Activity","Details"],
          ["1","Topic selection and planning","Topic and objectives finalised on 13-07-2026; quest
      ionnaire designed."],
          ["2","Survey","Questionnaire-based survey of 100 residents of Gudipallipadu."],
          ["3","Data analysis","Responses tabulated and converted to percentages to find awareness
       and health effects."],
          ["4","Report preparation","Findings, conclusion and recommendations written and reviewed
       with the Faculty Guide."]],
         [1.4*cm,4.6*cm,10.6*cm]),Spacer(1,6),
     P("<b>Sample size:</b> 100 residents. <b>Tool:</b> structured questionnaire (Yes/No and short
      -answer questions). "
       "<b>Analysis:</b> percentage analysis presented through tables and charts."),PageBreak()]

    # ---------- 4 Questionnaire ----------
    S+=[P("4. Questionnaire",h1),
     P("The questionnaire covered four areas. Residents were interviewed politely and without dist
      urbing their households."),
     tbl([["Section","Questions"],
          ["Demographic characteristics","Age, sex, occupation, education level, income level"],
          ["Health status","Do you suffer from any respiratory disease? Have you experienced healt
      h problems due to air pollution? "
                           "Have you visited a doctor or hospital for air pollution related issues
      ? (Yes/No)"],
          ["Economic activities","Primary source of income? Do you own agricultural land or livest
      ock? "
                                 "Have you experienced economic losses due to air pollution? (Yes/
      No)"],
          ["Awareness","Are you aware of the causes and effects of air pollution? Have you taken a
      ny measures to reduce exposure? "
                       "Do you think air pollution is a major problem in your area? (Yes/No)"]],[
      4.6*cm,12*cm]),
     Spacer(1,10),
     P("5. Data Analysis and Results",h1),
     P("The survey responses were analysed and are summarised below. All percentages are out of 10
      0 residents."),
     tbl([["Parameter","Result","Residents (out of 100)"],
          ["Aware of causes and effects of air pollution","75%","75"],
          ["Not aware","25%","25"],
          ["Reported health problems due to air pollution","40%","40"],
          ["Reported respiratory problems","40%","40"],
          ["Reported cardiovascular problems","20%","20"],
          ["Main income: agriculture","50%","50"],
          ["Main income: daily wage labour","30%","30"],
          ["Main income: other sources","20%","20"],
          ["Reported economic losses due to air pollution","20%","20"]],[8.6*cm,3.4*cm,4.6*cm]),
     PageBreak()]

    # ---------- Charts ----------
    S+=[P("5.1 Output: Charts of Survey Results",h1),
     img(os.path.join(CH,"c1.png"),7.6*cm),P("Figure 1: Awareness of causes and effects of air pol
      lution",cap),
     img(os.path.join(CH,"c2.png"),10.5*cm),P("Figure 2: Health effects reported by residents",cap
      ),
     PageBreak(),
     img(os.path.join(CH,"c3.png"),10.5*cm),P("Figure 3: Main source of income of surveyed residen
      ts",cap),
```

```
     img(os.path.join(CH,"c4.png"),13*cm),P("Figure 4: Summary of survey outcomes",cap),PageBreak(
      )]

    # ---------- 6-7 ----------
    S+=[P("6. Findings",h1),
     bullets(["<b>Awareness:</b> 75% of residents know the causes and effects of air pollution, wh
      ile 25% do not.",
              "<b>Health effects:</b> 40% report respiratory problems and 20% report cardiovascula
      r problems.",
              "<b>Livelihood:</b> Agriculture (50%) and daily wage labour (30%) are the main sourc
      es of income, "
              "so residents spend long hours outdoors.",
              "<b>Economic impact:</b> 20% of residents report economic losses due to air pollutio
      n.",
              "<b>Suggested measures:</b> Residents suggested promoting clean energy, improving wa
      ste management "
              "practices and increasing green cover."]),
     P("7. Discussion",h1),
     P("Although three out of four residents are aware of air pollution, a large share still repor
      t health problems. "
       "This suggests that awareness alone is not enough; practical measures are needed to reduce
      exposure at the source. "
       "Since many residents work in agriculture and as daily wage labourers, they are exposed to
      dust, smoke and "
       "burning of waste for long periods. The 25% of residents who are not aware of the causes an
      d effects show that "
       "awareness programmes should continue and reach every household."),
     P("8. Recommendations",h1),
     tbl([["Recommendation","Action"],
          ["Promote clean energy","Encourage solar, wind and biogas to reduce dependence on fossil
       fuels."],
          ["Improve waste management","Segregation, recycling and composting to reduce waste burni
      ng."],
          ["Increase green cover","Plant trees and develop green spaces to absorb pollutants."],
          ["Raise awareness","Continue campaigns, school programmes and community meetings."],
          ["Monitor air quality","Set up air quality monitoring to track pollutant levels and guid
      e decisions."]],[5*cm,11.6*cm]),
     PageBreak()]

    # ---------- 9-10 ----------
    S+=[P("9. Conclusion, Limitations and Future Scope",h1),
     P("<b>Conclusion.</b> The mini project shows that air pollution has a significant impact on h
      ealth and environment in "
       "Gudipallipadu. Awareness is fairly high (75%), but 40% of residents still report health pr
      oblems and 20% report "
       "economic losses. Promoting clean energy, improving waste management and increasing green c
      over can help "
       "reduce air pollution, along with continued awareness programmes."),
     P("<b>Limitations.</b> The study was limited to Gudipallipadu and a sample of 100 residents,
      and the results are based on "
       "self-reported responses. It may not be representative of other areas."),
     P("<b>Future scope.</b> Further studies can assess the impact of air pollution on specific he
      alth outcomes, evaluate "
       "the effectiveness of awareness programmes, and develop strategies to promote clean energy
      and better waste management."),
     P("10. References",h1),
     bullets(["World Health Organization (WHO): Air pollution and health fact sheets.",
              "Central Pollution Control Board (CPCB), Government of India: National Air Quality I
      ndex.",
              "AP State Council of Higher Education: Guidelines for the OJT Internship / Community
       Service Project.",
              "Primary data: questionnaire survey of 100 residents of Gudipallipadu, July 2026."])
      ,
     Spacer(1,1.5*cm),
     Table([[P("Signature of the Student",cen),P("Signature of the Faculty Guide",cen)]],colWidths
      =[8.3*cm,8.3*cm])]

    doc.build(S)
    print("PDF created: output/Mini_Project_Report_Air_Pollution.pdf")

build_pdf()
```

Output:

```
PDF created: output/Mini_Project_Report_Air_Pollution.pdf
```

Step 6: Download (Colab only)

In [6]:

```
# Optional: works only in Google Colab (on your own computer the PDF is already in the "output" fo
      lder)
try:
    from google.colab import files
    files.download("output/Mini_Project_Report_Air_Pollution.pdf")
except ImportError:
    print("Not running in Colab: open the 'output' folder to find Mini_Project_Report_Air_Pollutio
      n.pdf")
```

Output:

```
Not running in Colab: open the 'output' folder to find Mini_Project_Report_Air_Pollution.pdf
```